



TaC: An Anti-Caching Key-Value Store on Heterogeneous Memory Architectures

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In-Memory Key Value Stores

- **Fundamental storage layer**

- E.g. Redis, Memcached

- **Operation simple**

- Get, Set

- **Performance critical**

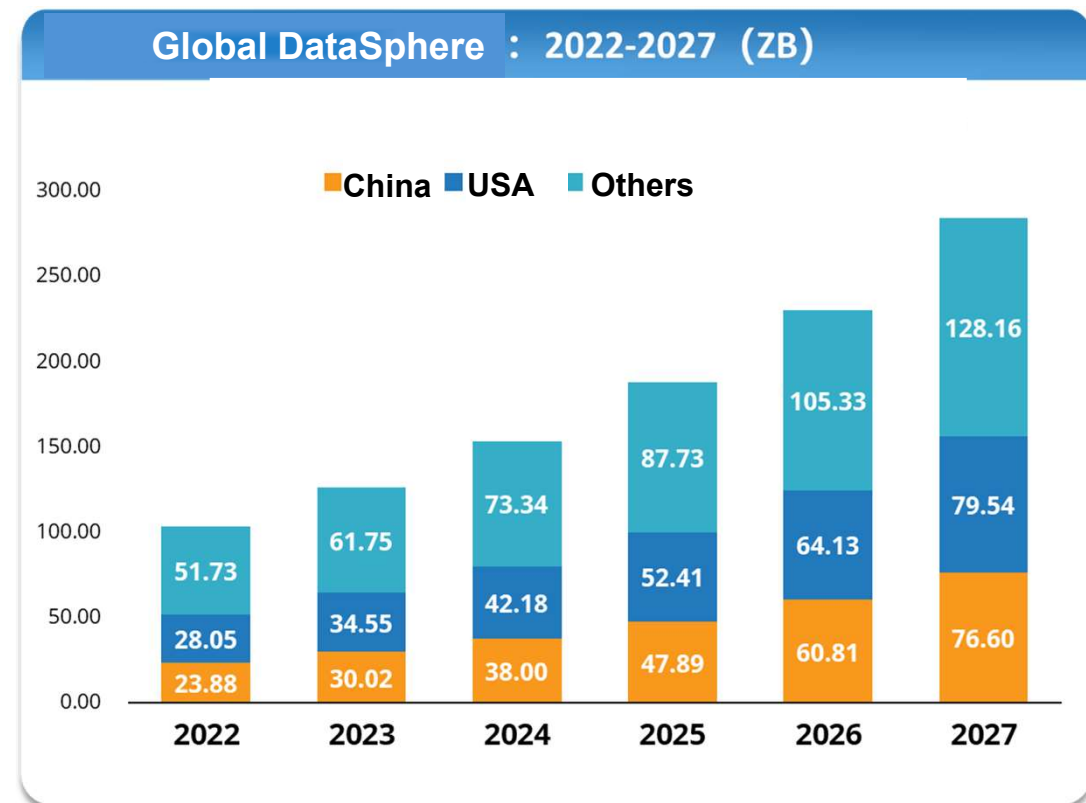


The Problem

- The increasing data volume
V.S.
- The memory scaling wall

The Increasing Data Volume

- Data volume remains increasing



Source: IDC Global DataSphere 2023

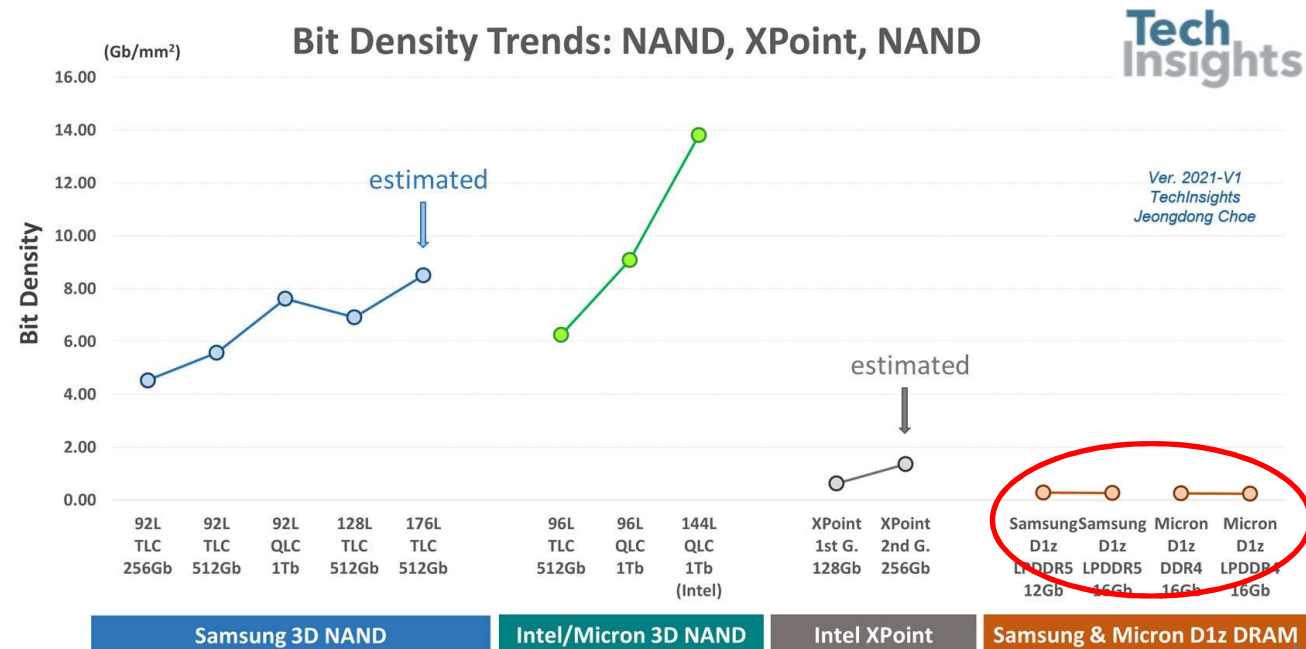
The Memory Scaling Challenge

- DRAM scaling wall

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- Scaling DRAM technologies to sub-20nm is challenging



The Problem

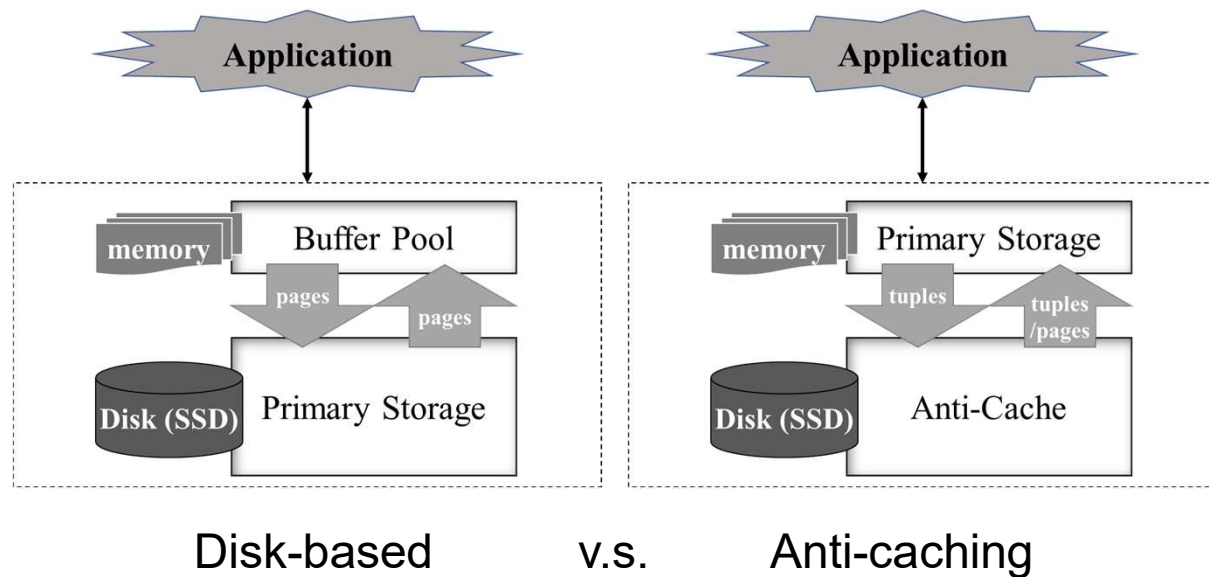
- **Growing imbalance between Memory demand and supply**
 - Increasing demand of memory in applications
 - DRAM capacity per core dropping by 30% biannually

The Problem

- **Growing imbalance between Memory demand and supply**
 - Increasing demand of memory in applications
 - DRAM capacity per core dropping by 30% biannually
- **Memory becomes more expensive compared to other devices**
 - 2TB M.2 SSD < \$500
 - 2TB DRAM ≈ \$20,000

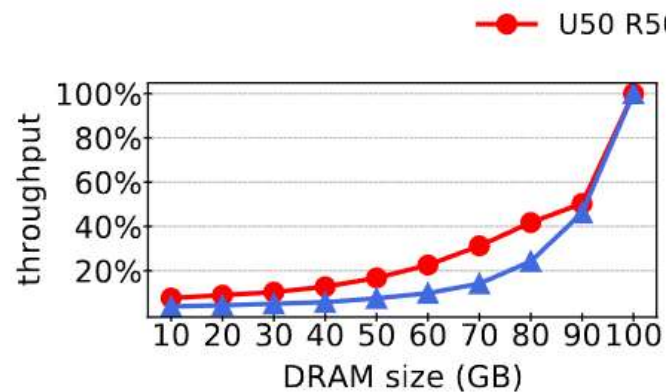
Anti-caching Architecture

- Expand the capacity while providing excellent performance

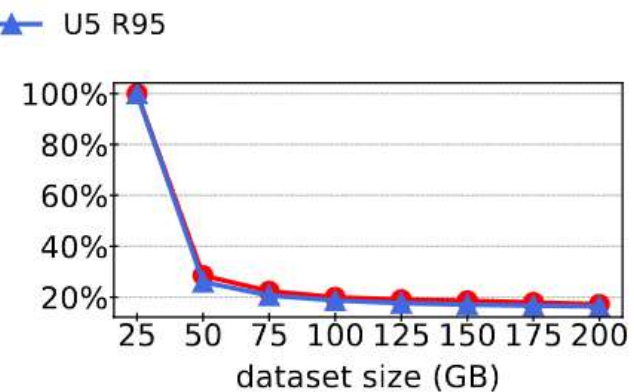


Anti-caching Architecture

- Performance diminishes when hot data surpasses the available memory size



(a) dataset = 100GB

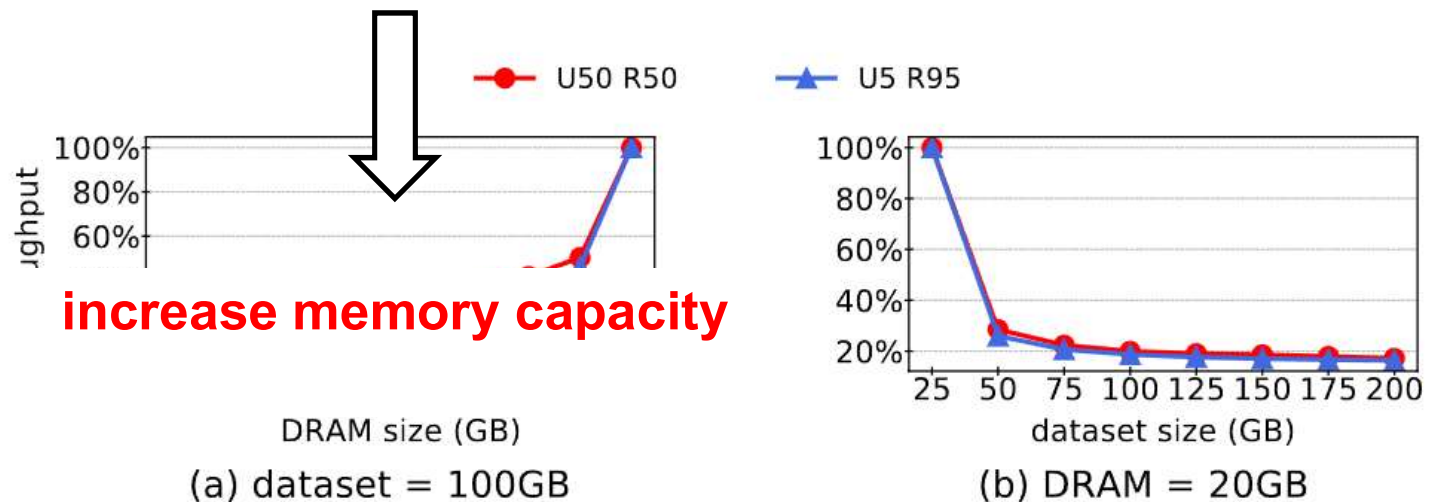


(b) DRAM = 20GB

Normalized performance of FASTER with different data volume and DRAM configuration

Anti-caching Architecture

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Normalized performance of FASTER with different data volume and DRAM configuration

Non-Volatile Memory

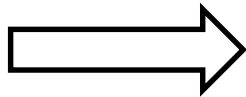
- Byte-addressable
- Larger capacity
- Lower prices

Table 1: Capacity and prices (\$/GB) of different memory.

Capacity	128GB	256GB	512GB
DDR5 DRAM DIMM [12]	11.3	12.5	-
Intel Optane PMem [32]	8.6	8.4	8.2

Non-Volatile Memory

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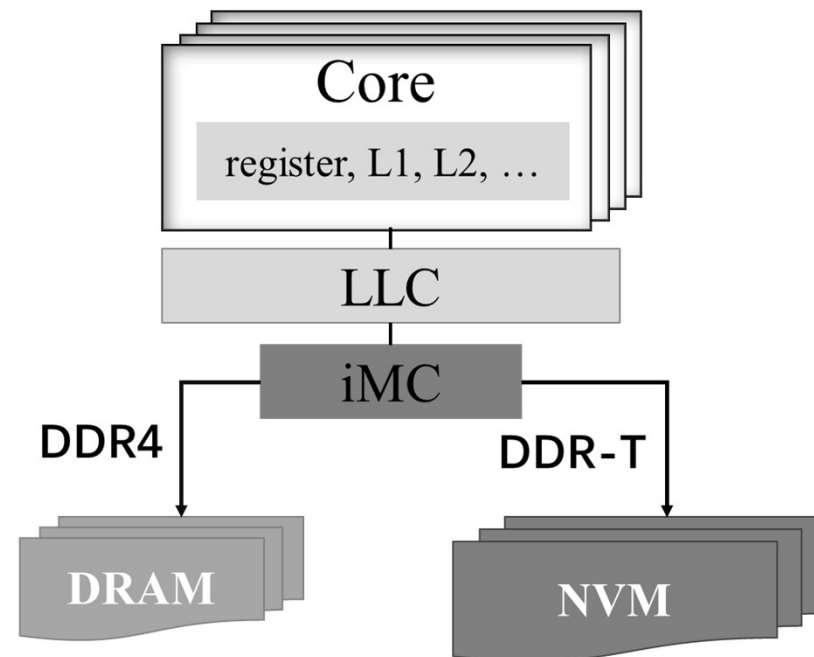
**Leverage NVM to expand the
memory in anti-caching
Key-Value stores**

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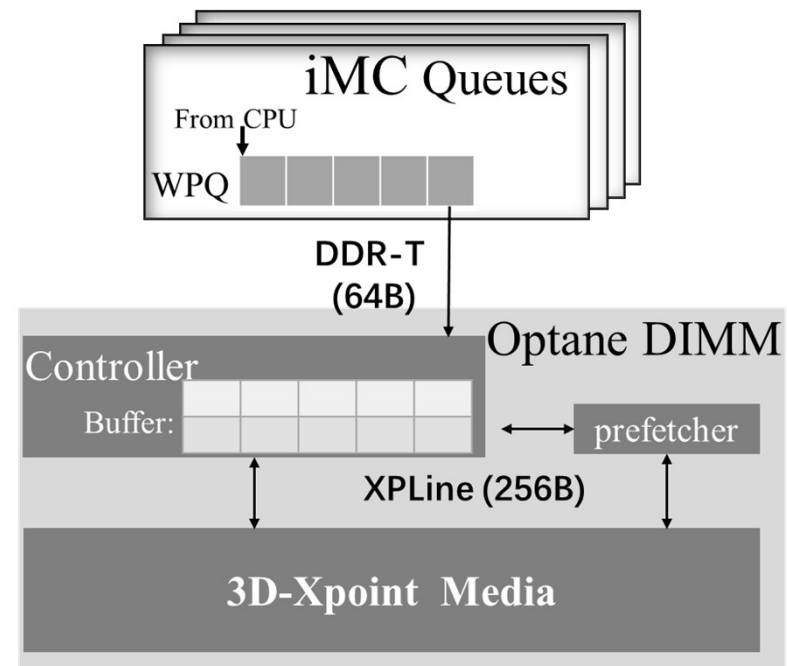
Non-Volatile Memory

- NVDIMM-P



Non-Volatile Memory

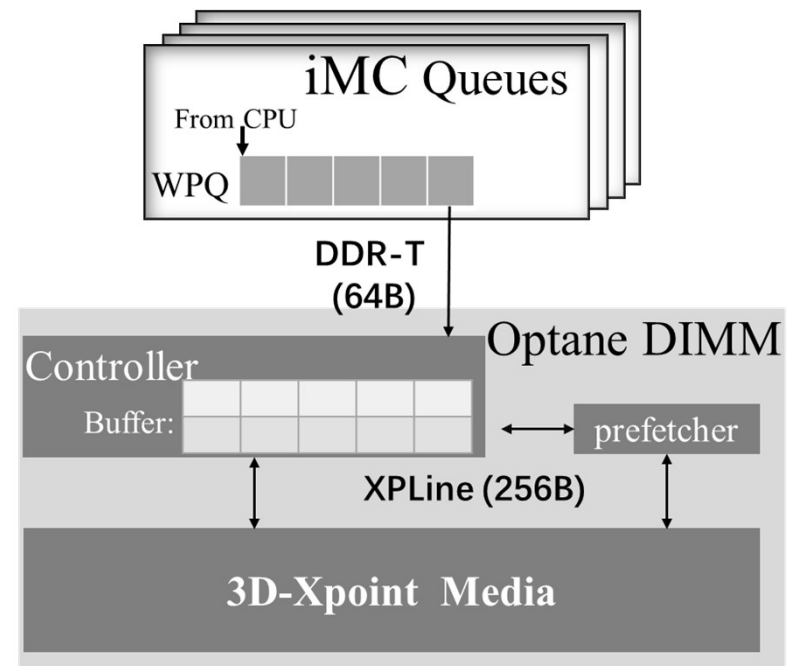
- Performance characteristics



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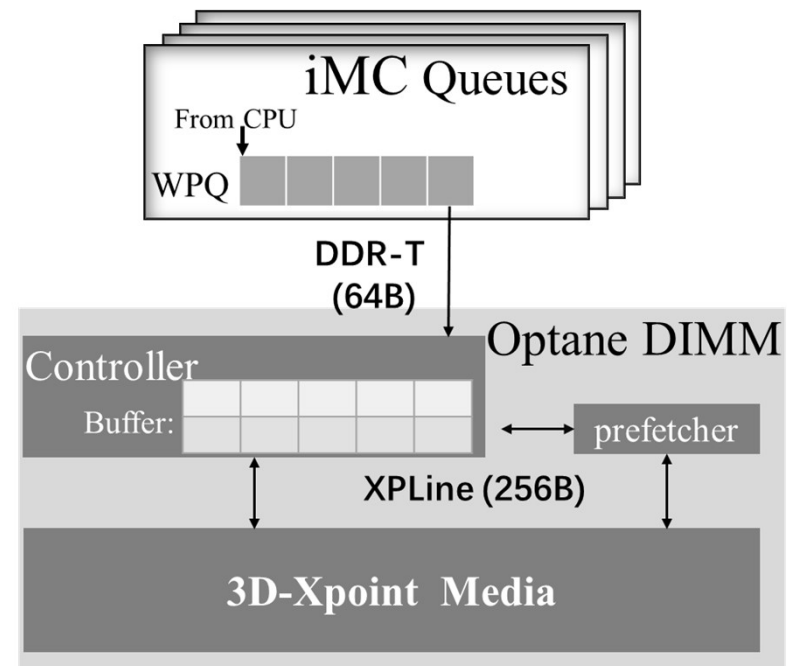
- p1: Inferior performance
- p2: Read-write asymmetry



Non-Volatile Memory

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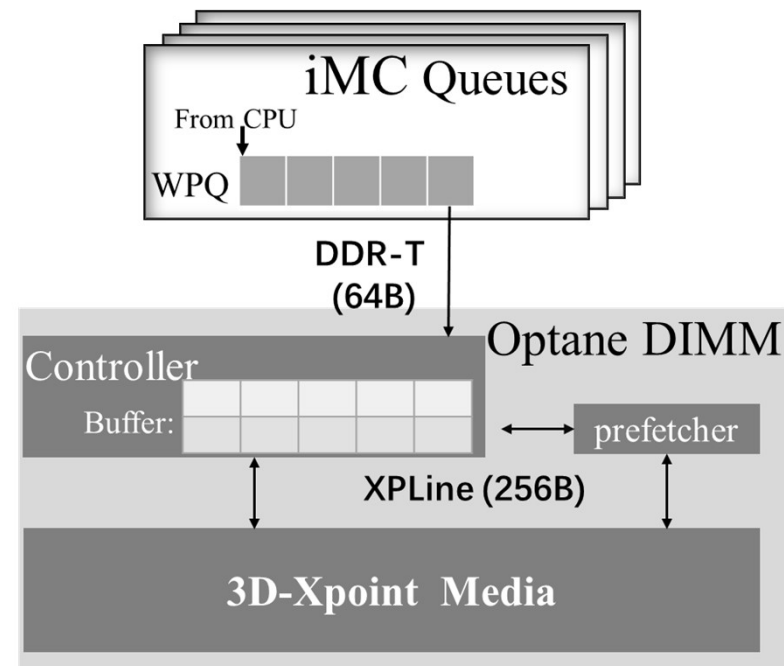
- p1: Inferior performance
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- p3: Inferior performance on small and random accesses



Non-Volatile Memory

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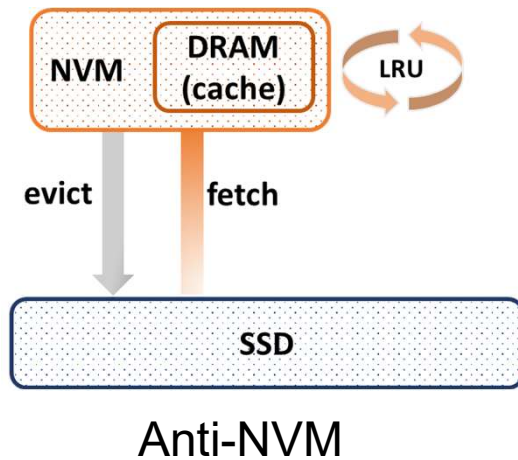
- p1: Inferior performance
- p2: Read-write asymmetry
- p3: Inferior performance on small and random accesses
- p4: Limited concurrency
- p5: Interference with DRAM



Anti-Caching with NVM

- **Potential choices**

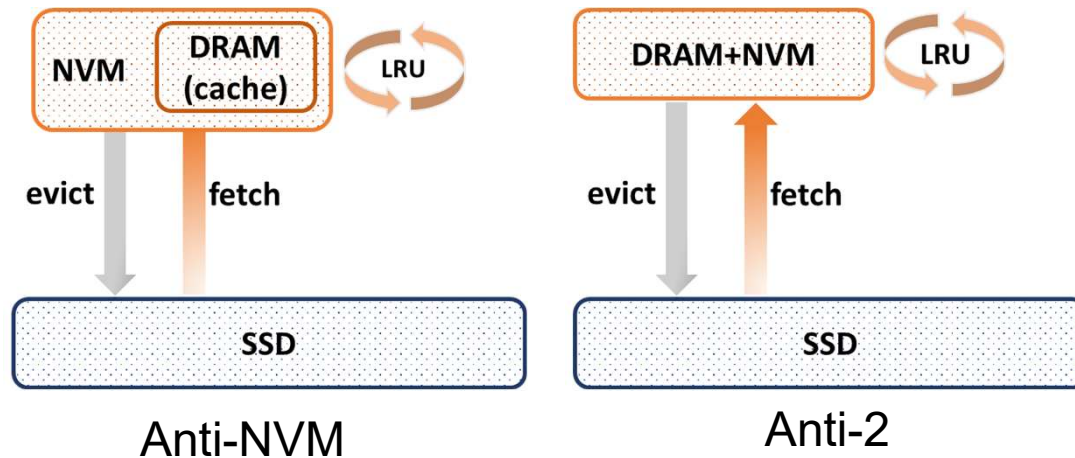
- Anti-NVM: Replace DRAM with NVM, similar to “Memory Mode”



Anti-Caching with NVM

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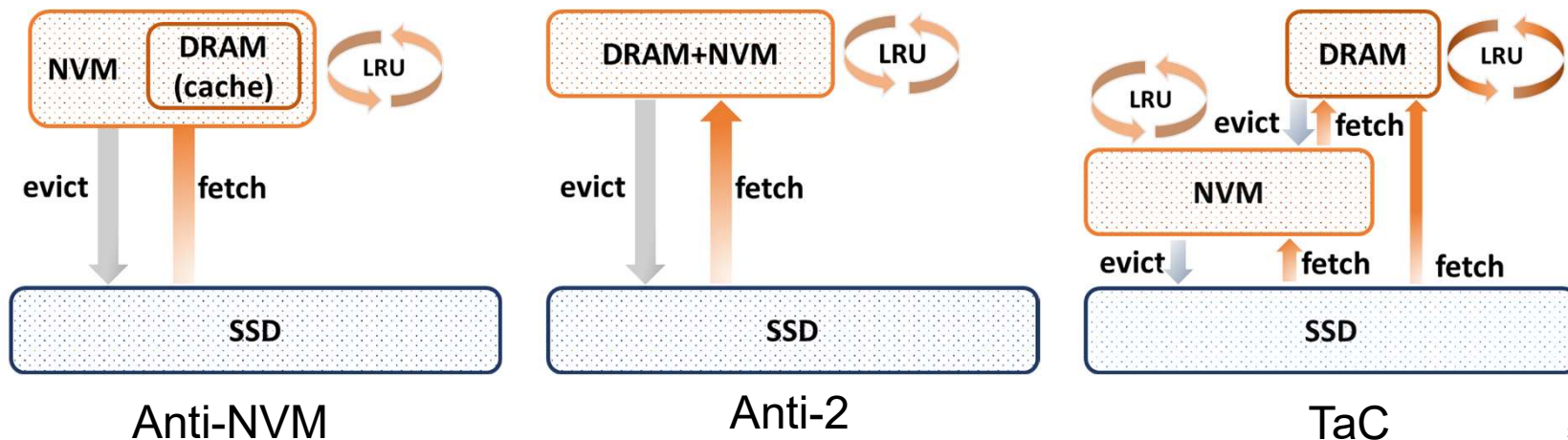
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- Anti-2: Utilize NVM in the same way as DRAM



Anti-Caching with NVM

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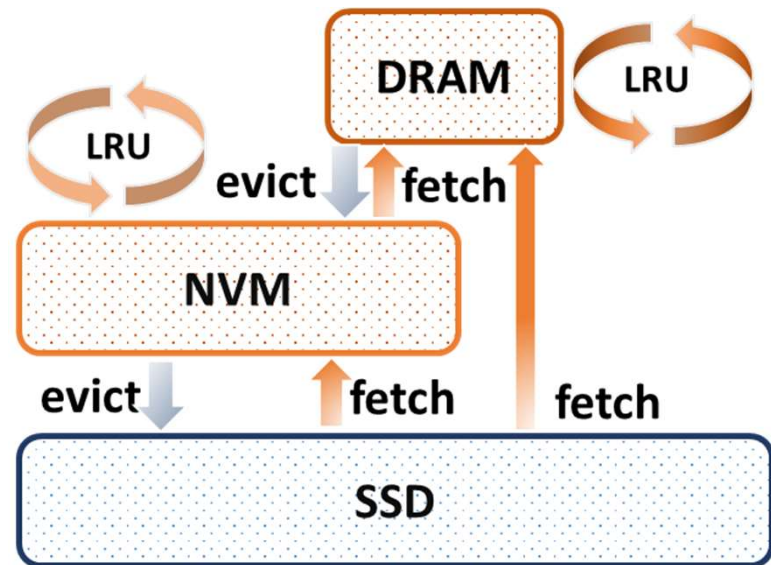
- Anti-NVM: Replace DRAM with NVM, similar to “Memory Mode”
- Anti-2: Utilize NVM in the same way as DRAM
- TaC: Three-tier anti-Caching



Three-tier anti-Caching

- Strengths

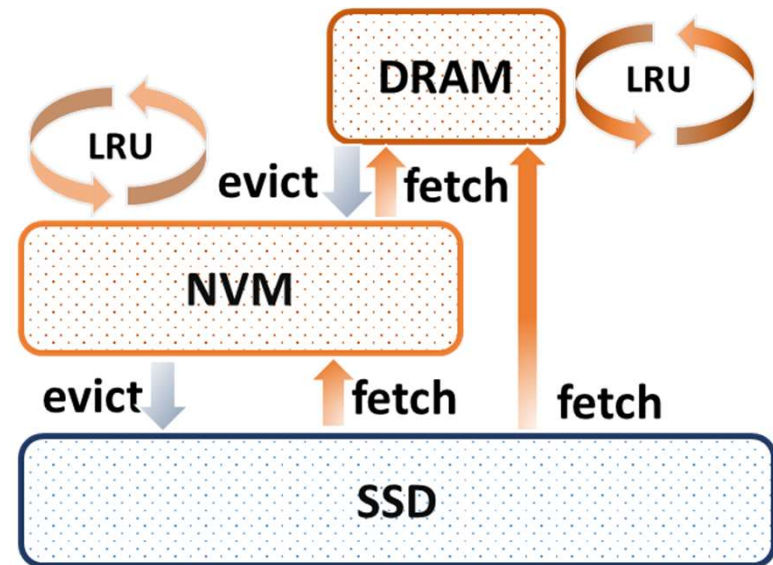
- Leveraging DRAM Strengths (p1)



Three-tier anti-Caching

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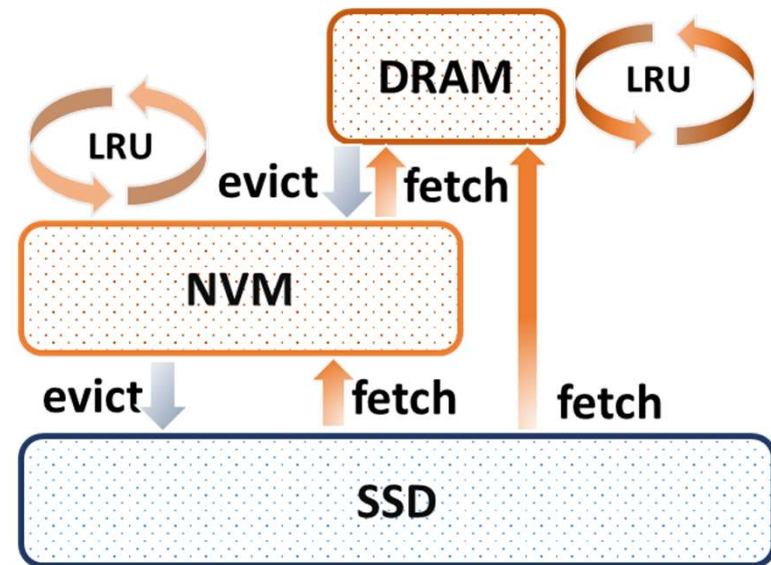
- Leveraging DRAM Strengths (p1)
- Hot Data Retention



Three-tier anti-Caching

- **Strengths**

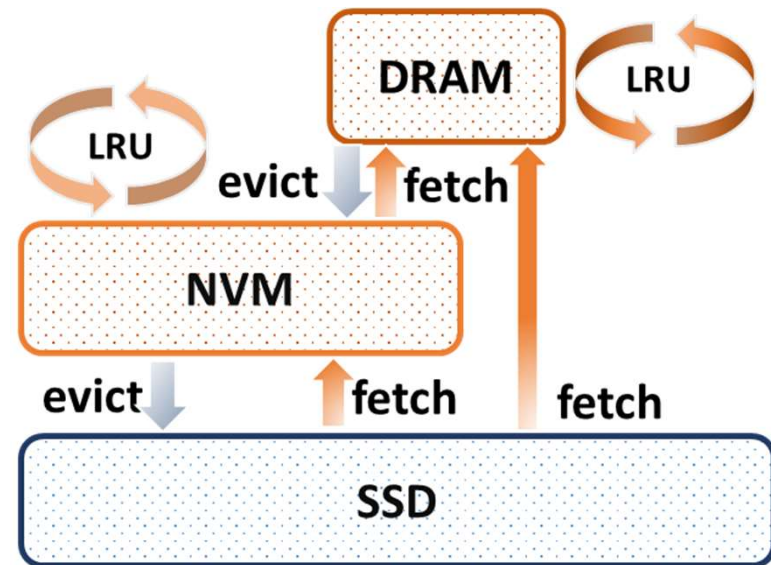
- Leveraging DRAM Strengths (p1)
- Hot Data Retention
- Controlled Writing to NVM (p4, p5)



Three-tier anti-Caching

- **Challenges**

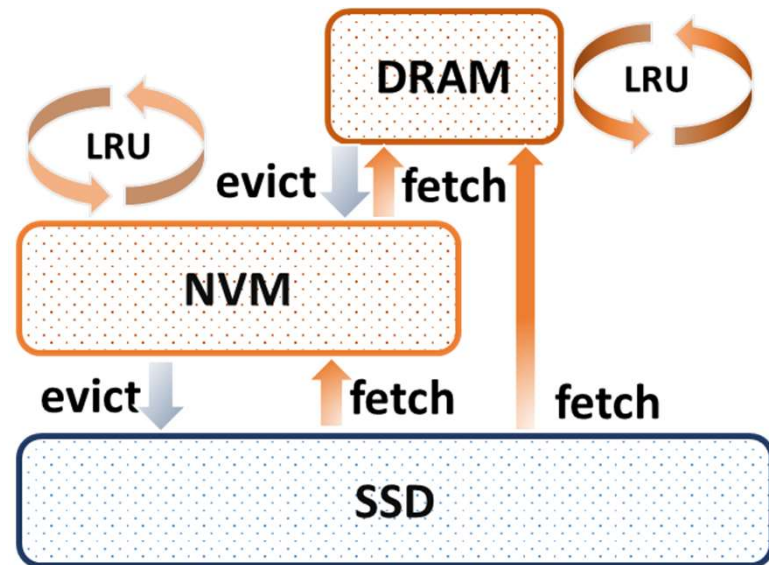
- More complex data swapping paths



Three-tier anti-Caching

- **Challenges**

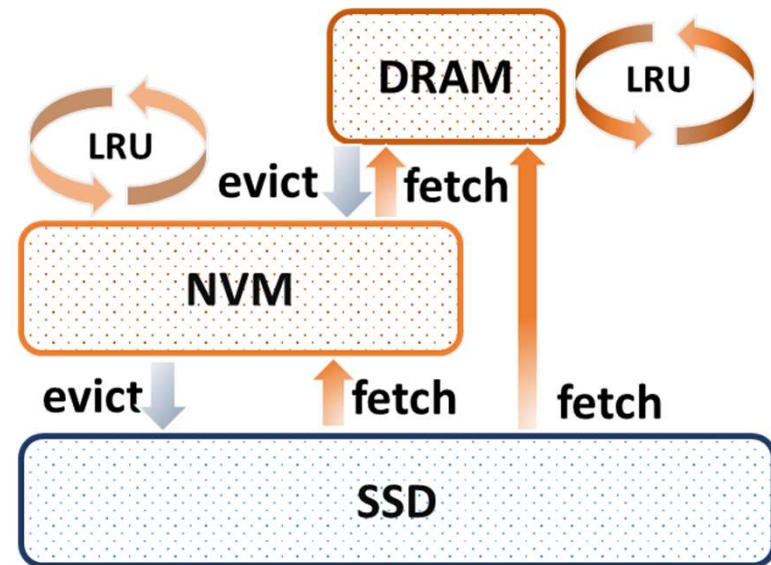
- More complex data swapping paths
- Multi-level data classification



Three-tier anti-Caching

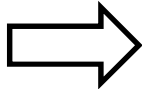
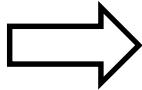
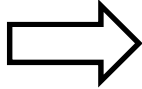
- **Challenges**

- More complex data swapping paths
- Multi-level data classification
- NVM-specific characteristics



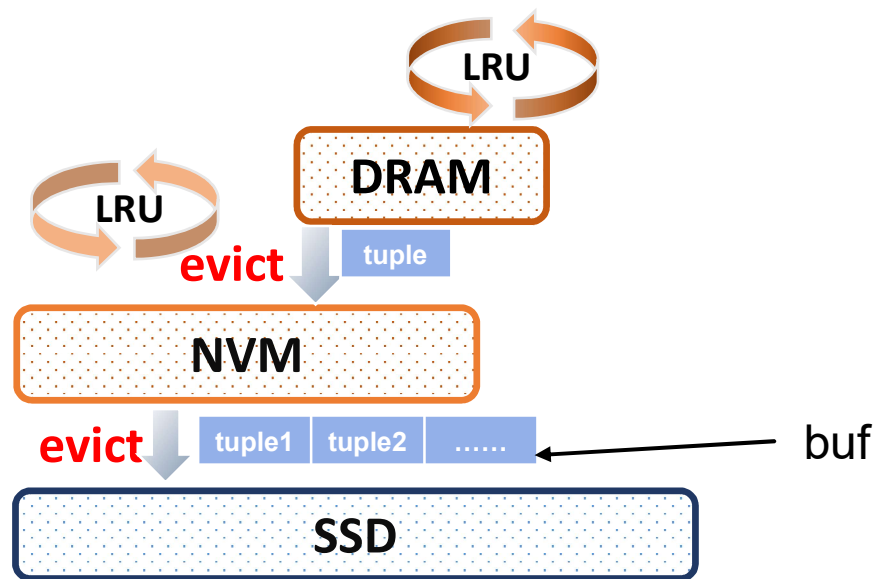
Three-tier anti-Caching

- **Challenges**

- | | | |
|------------------------------------|---|--------------------------------|
| ▪ More complex data swapping paths |  | Hybrid data swapping |
| ▪ Multi-level data classification |  | Lazy LRU |
| ▪ NVM-specific characteristics |  | NVM-optimized data arrangement |

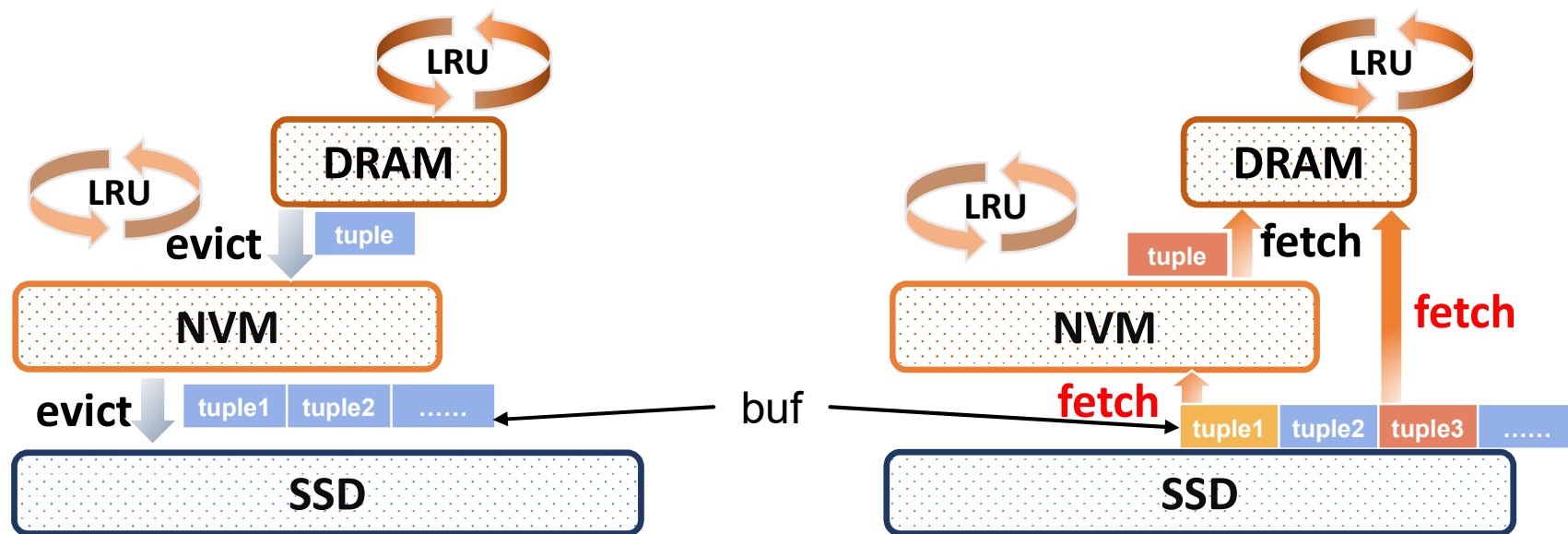
Three-tier anti-Caching

- Hybrid data swapping



Three-tier anti-Caching

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Three-tier anti-Caching

- **Lazy LRU**

- $time$, the previous access time
- $readCount$, the recent access frequency

Table 2: Temperature classes of tuples.

$t.readCount$	Temperature	Action
$> THR_{hot}$	Hot	Fetch from SSD or NVM to DRAM; Move atop the LRU list if in DRAM.
$> THR_{warm}$	Warm	Fetch from SSD to NVM; Move atop the LRU list if in NVM.
Others	Cold	Become ready for eviction.

Three-tier anti-Caching

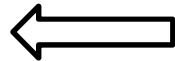
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- Multi-level data classification
- asynchronous updates of LRU lists



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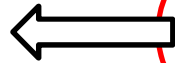
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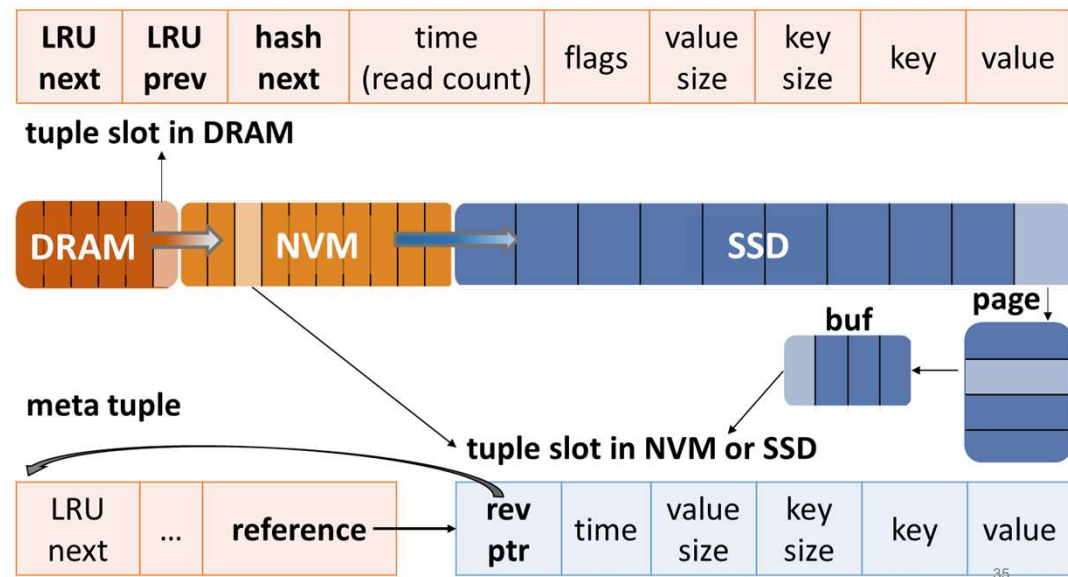
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- Sample-based thresholds



Three-tier anti-Caching

- **NVM-optimized data arrangement based on Memcached**
 - Manage NVM space at tuple-level like DRAM
 - Maintain the metadata in DRAM



Evaluation

- **Compared systems**

- Anti-NVM
- Anti-2 & FASTER-NVM
- Spitfire & PRISM

- **Set-Up**

- 4GB DRAM
- 32GB NVM

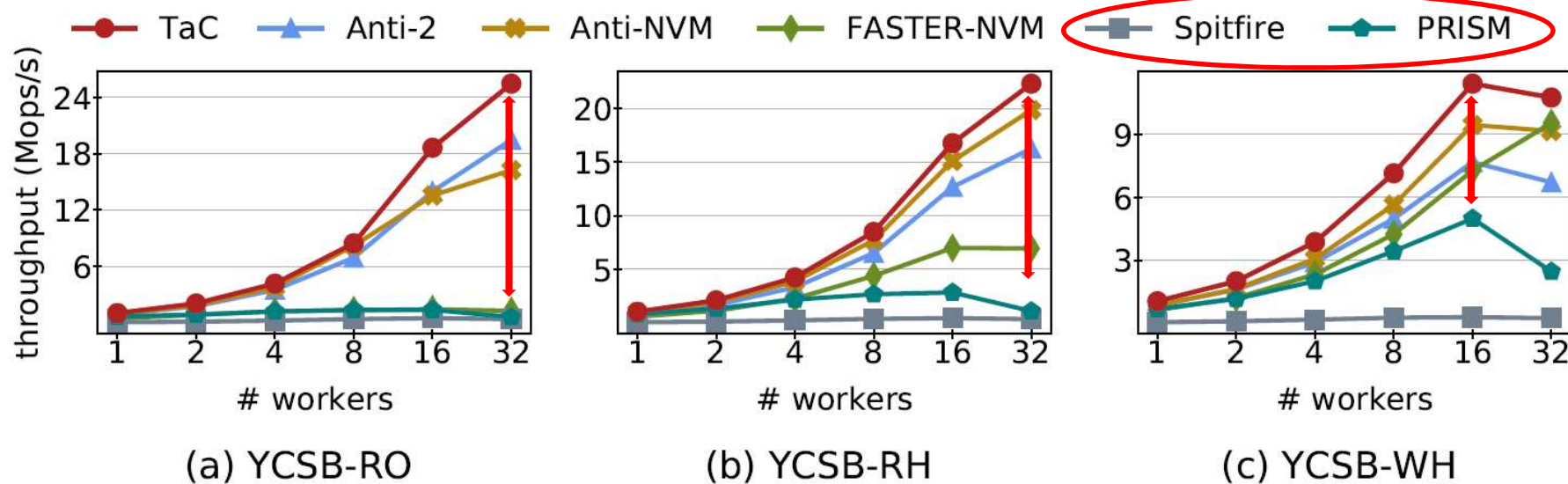
- **Workloads**

- Read-Only (YSCB-RO): 100% reads
- Read-Heavy (YCSB-RH): 95% reads, 5% updates
- Write-Heavy (YCSB-WH): 50% reads, 50% updates

Evaluation

• Observations

- The anti-caching architecture outperforms the caching architecture regarding the throughput



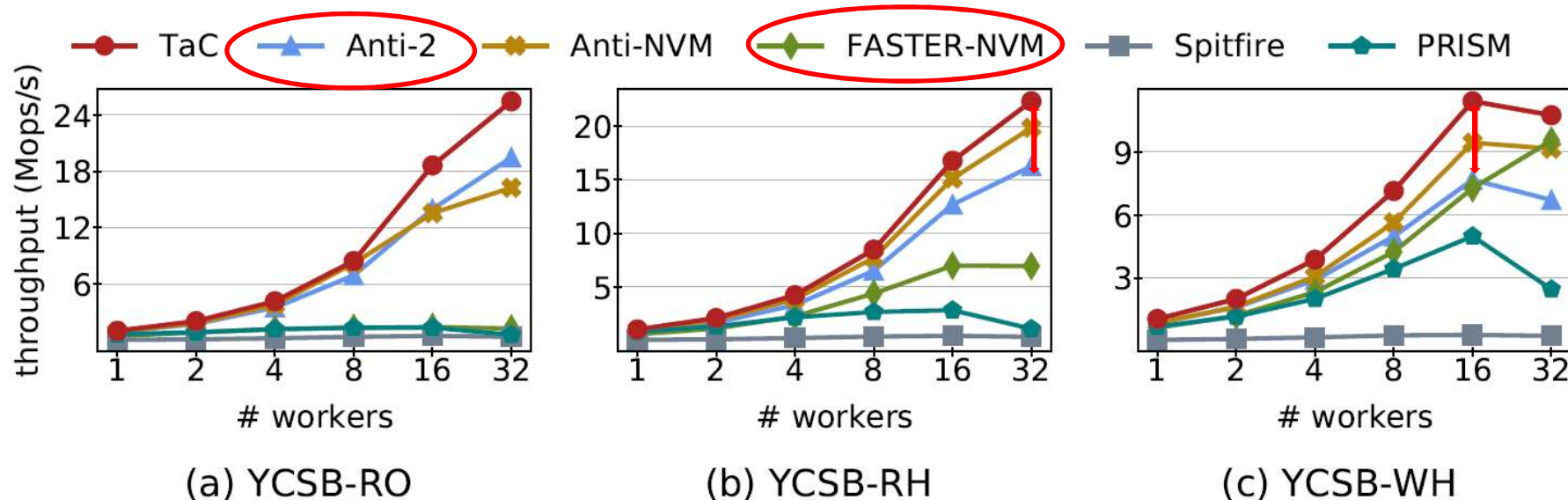
Evaluation

• Observations

- Distinguishing between NVM and DRAM enhances system performance by improving DRAM hit rates and NVM accessing efficiency

Table 3: Memory hit rates on the YCSB-RH workload.

	TaC	Anti-2	Anti-NVM
DRAM	61.4%	9.1%	59.8%
NVM	21.0%	76.3%	18.8%



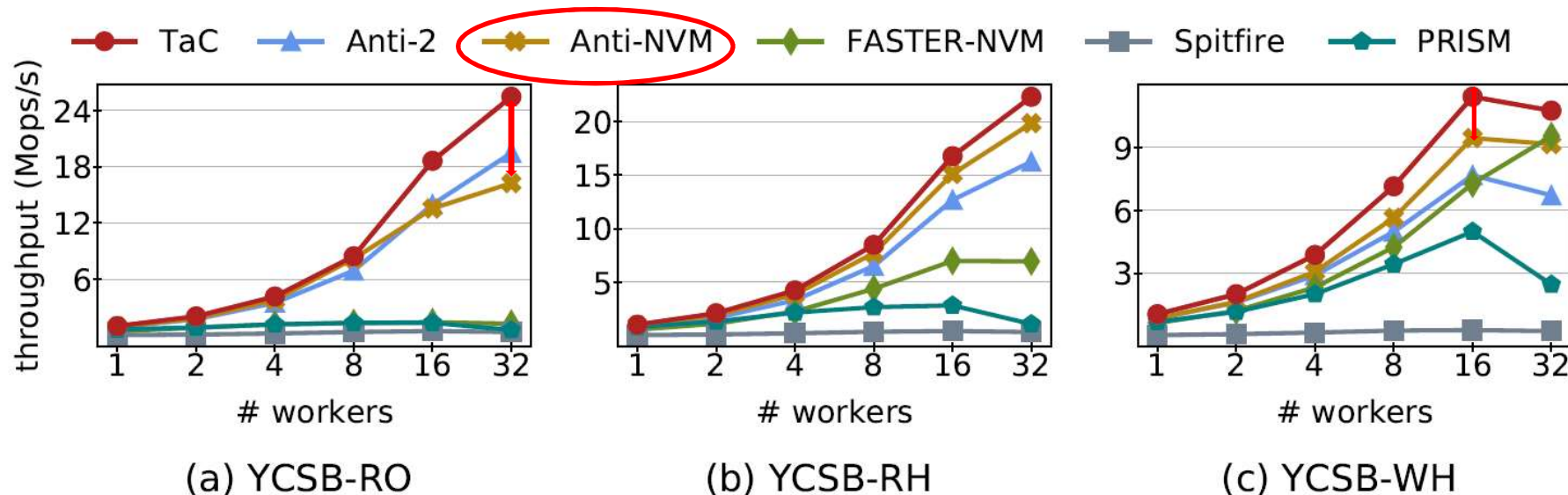
Evaluation

• Observations

- The three-tier architecture outperforms the "Memory Mode" architecture due to better memory utilization and reduced data synchronization cost

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Evaluation

- **More experiments**

- Experiments on varying DRAM and NVM sizes
- Experiments on varying data volume
- Optimization Impact Analysis
 - Partitioned memory management
 - Hybrid data swapping
 - Lazy LRU
-

Thanks!